LECTURE 14 (153-248)

All models so far: Mean Models

$$y_t = \underbrace{\beta_0 + \beta_1(t-1) + \sum_{j=2}^{n-1} \beta_j \, \text{ReLU}(t-j)}_{\mu_t} + \varepsilon_t$$

$\varepsilon_t \overset{iid}{\sim} N(0, \sigma^2)$

$$y_t = \mu_t + \varepsilon_t$$

$$y_t \overset{\text{independent}}{\sim} N(\mu_t, \sigma^2)$$

Estimation: $\sum_{t=1}^{n} (y_t - \mu_t)^2$

$$\sum_{t=1}^{n} \left[y_t - \underbrace{\beta_0 - \beta_1(t-1) - \sum_{j=2}^{n-1} \beta_j \, \text{ReLU}(t-j)}_{\mu_t} \right]^2$$

$$+ \lambda \sum_{j=2}^{n-1} \beta_j^2 \quad \text{or} \quad \lambda \sum_{j=2}^{n-1} |\beta_j|$$

Rewrite:

$$\sum_{t=1}^{n} (y_t - \mu_t)^2$$

$$\mu_t = \beta_0 + \beta_1(t-1) + \beta_2 \, \text{ReLU}(t-2) + \cdots + \beta_{n-1} \, \text{ReLU}(t-(n-1))$$

$t = 1, \ldots, n$

$$\beta_0 = \mu_1$$

$$\beta_1 = \mu_2 - \mu_1$$

$$\beta_2 = (\mu_3 - \mu_2) - (\mu_2 - \mu_1)$$

$$\beta_j = (\mu_{j+1} - \mu_j) - (\mu_j - \mu_{j-1})$$

$$\beta_{n-1} = (\mu_n - \mu_{n-1}) - (\mu_{n-1} - \mu_{n-2})$$

So our estimate is

$$\sum_{t=1}^{n} (y_t - \mu_t)^2 + \lambda \sum_{t=2}^{n-1} \left[(\mu_{t+1} - \mu_t) - (\mu_t - \mu_{t-1}) \right]^2$$

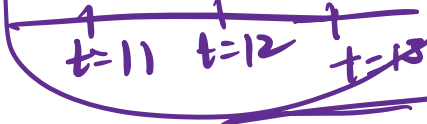
$$y_t \overset{ind}{\sim} N(\mu_t, \sigma^2)$$

$\mu_{t+1} - \mu_t \approx \mu_t - \mu_{t-1}$ in an average sense

Variance Modeling (Spectrum Model)

Variance Model

$$y_t \overset{ind}{\sim} N(0, \tau_t^2)$$



$y_t \sim N(\mu_t, \sigma^2)$

Likelihood:

$$\prod_{t=1}^{n} \left[\frac{1}{\sqrt{2\pi}\,\tau_t} \exp\left(-\frac{y_t^2}{2\tau_t^2} \right) \right]$$

$\Rightarrow$ Sufficient Statistic in this model is y_t^2.

Model: $y_t^2 \overset{ind}{\sim} \tau_t^2 \chi_1^2$ (i.e, $\frac{y_t^2}{\tau_t^2} \overset{iid}{\sim} \chi_1^2$)

Check that likelihood with $\tau_t^2 \chi_1^2$ coincides with likelihood based on $N(0, \tau_t^2)$ (after dropping constants)

$$\prod_{t=1}^{n} \frac{1}{\tau_t} \exp\left(-\frac{y_t^2}{2\tau_t^2}\right)$$

Log-likelihood

$$= \sum_{t=1}^{n} \left[-\frac{y_t^2}{2\tau_t^2} - \log \tau_t \right]$$

Negative log-likelihood

$$= \sum_{t=1}^{n} \left[\frac{y_t^2}{2\tau_t^2} + \log \tau_t \right]$$

Differentiate w.r.t τ_t:

$$-y_t^2 \tau_t^{-3} + \frac{1}{\tau_t} = 0$$

$$\Rightarrow \tau_t = |y_t| \text{ for each } t = 1, \ldots, n.$$

$\longrightarrow$ CORRESPONDS TO OVERFITTING.

We shall use regularized estimation.

Define $\boxed{\alpha_t := \log \tau_t}$

Negative log-likelihood

$$= \sum_{t=1}^{n} \left[\frac{y_t^2}{2} e^{-2\alpha_t} + \alpha_t \right]$$

Estimate $(\alpha_1 \dots \alpha_n)$ by

$$\underset{\alpha_1 \dots \alpha_n}{\text{Minimize}} \left\{ \sum_{t=1}^{n} \left[\frac{y_t^2}{2} e^{-2\alpha_t} + \alpha_t \right] + \lambda \sum_{t=2}^{n-1} \left[(\alpha_{t+1} - \alpha_t) - (\alpha_t - \alpha_{t-1}) \right]^2 \right\}$$

$$\text{or } \lambda \sum_{t=2}^{n-1} \left| \underbrace{(\alpha_{t+1} - \alpha_t) - (\alpha_t - \alpha_{t-1})}_{\alpha_{t+1} - 2\alpha_t + \alpha_{t-1}} \right|$$

Spectrum Model

Data: $y_0 \dots y_{n-1}$

DFT: $b_0 \dots b_{n-1}$

$$b_j = \sum_{t=0}^{n-1} y_t \exp\left(-\frac{2\pi i j t}{n} \right)$$

$$\rightarrow u^j = \left(\exp\left(\frac{2\pi i j t}{n} \right),\ t = 0, 1, \dots, n-1 \right)$$

$$\boxed{b_j = \langle y, u^j \rangle}$$

$$\langle x, y \rangle = \sum_j x_j \bar{y}_j$$

$$\mathrm{Re}(b_j) = \sum_{t=0}^{n-1} y_t \cos\left(2\pi \frac{j}{n} t\right)$$

$$\mathrm{Im}(b_j) = -\sum_{t=0}^{n-1} y_t \sin\left(2\pi \frac{j}{n} t\right)$$

$$b_0 = \sum_{t=0}^{n-1} y_t \longrightarrow \text{not very useful}$$

$$b_{n-j} = \sum_{t=0}^{n-1} y_t \exp\left(-\frac{2\pi i (n-j)}{n} t\right)$$

$$= \sum_{t=0}^{n-1} y_t \exp\left(-2\pi i \left(1 - \frac{j}{n}\right) t\right)$$

$$= \sum_{t=0}^{n-1} y_t \underbrace{\exp(-2\pi i t)}_{=1} \exp\left(+2\pi i \frac{j}{n} t\right)$$

$$= \sum_{t=0}^{n-1} y_t \exp\left(2\pi i \frac{j}{n} t\right) = \overline{b_j}$$

$$b_{n-j} = \overline{b_j}$$

Data: $y_0, y_1, \ldots, y_{n-1}$

$n = 7$

$y_0, y_1, y_2, y_3, y_4, y_5, y_6$

DFT: $b_0, b_1, b_2, b_3, \overline{b_3}, \overline{b_2}, \overline{b_1}$

$b_4 = \overline{b_3}$

b_0 ↓ real sum of data

$n = 8$

$b_0, b_1, b_2, b_3, b_4, \overline{b_3}, \overline{b_2}, \overline{b_1}$

b_4 ↑

$b_{8-j} = \overline{b_j}$

$j = 4$ $\quad b_4 = \overline{b_4}$

Assume n is odd

$m = \frac{n-1}{2}$ is an integer

$b_1, \ldots, b_m$ → useful part of the DFT

Model: Variance model for DFT

$j = 1, \ldots, m$

$\text{Re}(b_j), \text{Im}(b_j) \overset{iid}{\sim} N(0, \gamma_j^2)$ $\quad j = 1, \ldots, m$

Independence across j.

$y_t \sim N(0, \tau_t^2)$

$\tau_t = |y_t|$

Periodogram

$$I\left(\frac{j}{n}\right) := \frac{|b_j|^2}{n} = \frac{(Re(b_j))^2 + (Im(b_j))^2}{n}$$

$$\overset{ind}{\sim} \frac{\gamma_j^2}{n} \chi_2^2$$

$$I\left(\frac{j}{n}\right) \overset{ind}{\sim} \frac{\gamma_j^2}{n} \chi_2^2$$

Likelihood: $b_1 \dots b_m$

$$Re(b_j) \overset{iid}{\sim} N(0, \gamma_j^2)$$

$$Im(b_j)$$

$$\prod_{j=1}^{m} \left[\frac{1}{\sqrt{2\pi}\,\gamma_j} \exp\left(-\frac{(Re(b_j))^2}{2\gamma_j^2} \right) \right.$$

$$\left. \frac{1}{\sqrt{2\pi}\,\gamma_j} \exp\left(-\frac{(Im(b_j))^2}{2\gamma_j^2} \right) \right]$$

$$\propto \prod_{j=1}^{m} \gamma_j^{-2} \exp\left(-\frac{(Re(b_j))^2 + (Im(b_j))^2}{2\gamma_j^2} \right)$$

$$= \prod_{j=1}^{m} \gamma_j^{-2} \exp\left(-\frac{|b_j|^2}{2\gamma_j^2} \right)$$

$I\left(\frac{j}{n}\right) = \frac{|b_j|^2}{n}$

$$\text{likelihood} = \prod_{j=1}^{m} \gamma_j^{-2} \exp\left(-\frac{I(j/n) \times n}{2\gamma_j^2}\right)$$

$$\log \text{lik} = \sum \left[-2\log \gamma_j - \frac{n}{2\gamma_j^2} I(j/n)\right]$$

$$\text{neg log lik} = \sum \left[2\log \gamma_j + \frac{n}{2\gamma_j^2} I(j/n)\right]$$

$$\alpha_j = \log \gamma_j$$

$$\sum \left[2\alpha_j + \frac{n}{2} I(j/n)\, e^{-2\alpha_j}\right]$$

derivatives: $2 + \frac{n}{2} I(j/n)\, e^{-2\alpha_j}(-2)$

$= 0$

$$\hat{\alpha}_j = \log \sqrt{\frac{n I(j/n)}{2}}$$ → maximizes

Model: $I(j/n) \sim \frac{\gamma_j^2}{n} \chi_2^2$

$$\mathbb{E}\left(I(j/n)\right) = \frac{\gamma_j^2}{n} \times 2 = \frac{2}{n} e^{2\alpha_j}$$

$$\gamma_j = e^{\alpha_j}$$

$$\frac{2}{n} e^{2\hat{\alpha}_j} = I(j/n)$$

Estimate:

$$\sum \left[2\alpha_j + \frac{n I(j/n)}{2} e^{-2\alpha_j} \right] + \lambda \sum_{j=2}^{m-1} \left(\alpha_{j+1} - 2\alpha_j + \alpha_{j-1} \right)^2$$

negative log-likelihood

$\alpha_1 \dots \alpha_m$

Periodogram

$I\left(\frac{j}{n}\right)$, $j = 1 \dots m$